

# BEACON

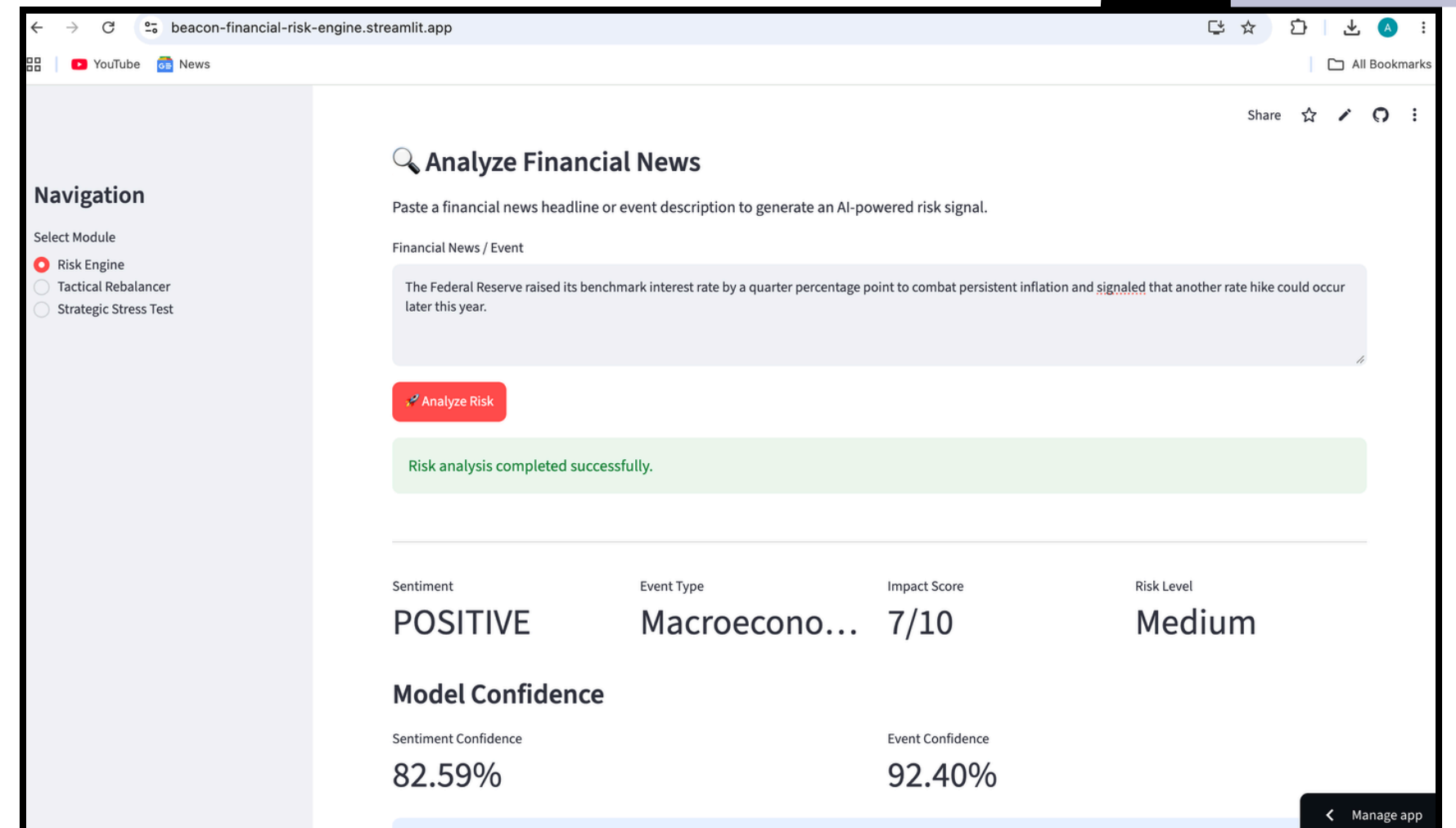
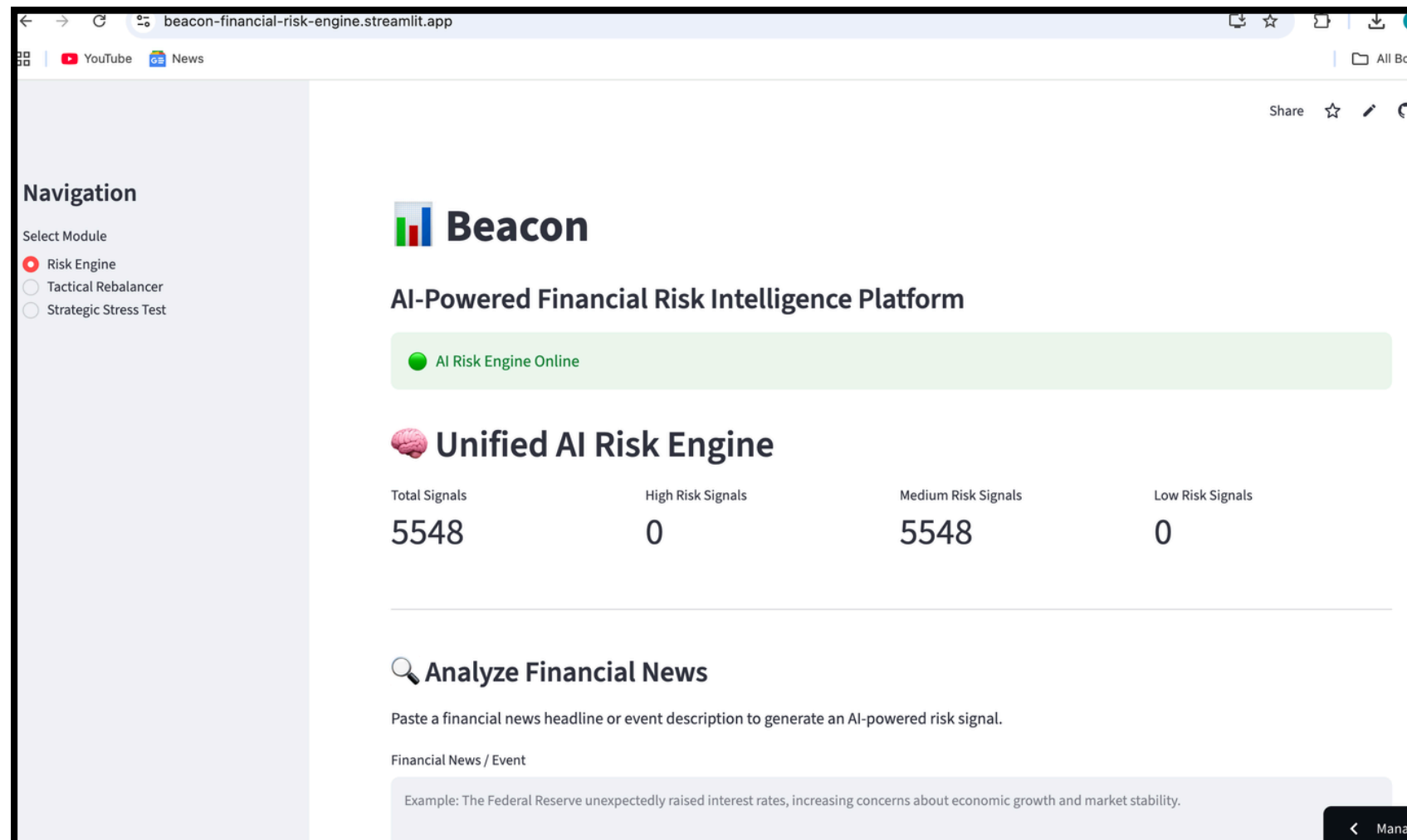
## AI-Powered Financial Risk Intelligence Platform

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Turning unstructured financial information into actionable risk intelligence.



# From Financial Noise to Actionable Risk Signals

## Problem

- Massive volumes of unstructured news and social media—too noisy for manual monitoring.
- Manual analysis is slow, subjective, and hard to scale across assets.
- Raw sentiment alone rarely maps cleanly to portfolio decisions.

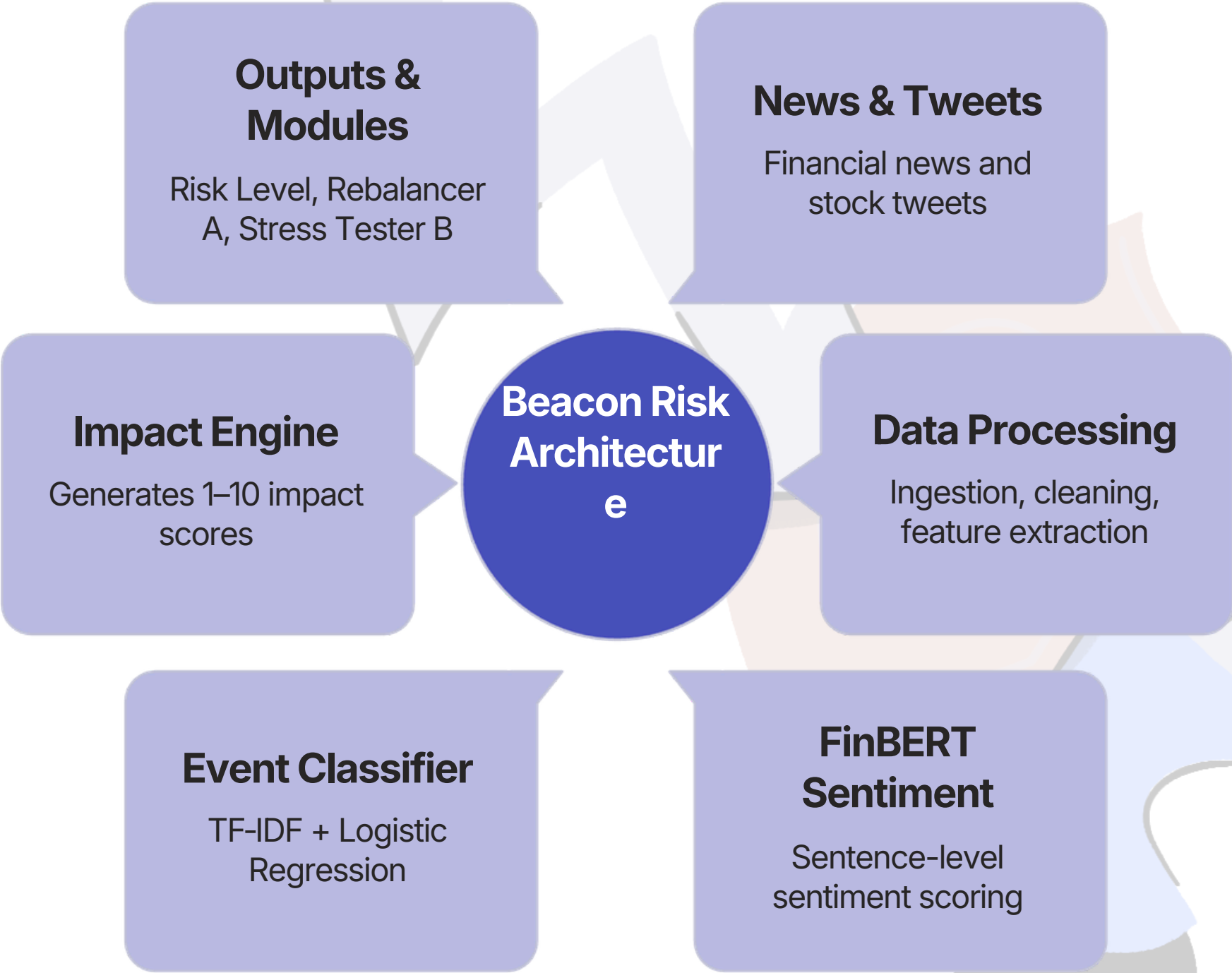
## Approach (pipeline)

News + Social Data → NLP Risk Engine → Sentiment + Event + Impact (1–10) → Actionable Risk Signal → Portfolio Decisions

Output schema: Sentiment | Event | Impact (1–10) | Risk Level — standardized for downstream automation.



# Beacon — End-to-End Architecture



Tech stack: Python · pandas · scikit-learn · FinBERT · Streamlit · Plotly



# Implementation Highlights

## Risk Engine

FinBERT → financial sentiment. TF-IDF + Logistic Regression → event classification. Impact Engine → estimated 1–10 impact score. Final mapping → Low / Medium / High risk.

## Module A — Tactical Rebalancer

Processed ~80K tweets across 25 stocks. Daily sentiment aggregation drives dynamic portfolio weighting for tactical allocation experiments.

## Module B — Strategic Stress Tester

Simulated a 100-asset synthetic portfolio under geopolitical triggers and interest-rate shocks to quantify portfolio impact before/after events.

## Design Principle

Used the best-performing model per subtask rather than a single all-purpose model — pragmatic, modular, and interpretable.

# Key Results — Performance & Validation

## NLP Performance

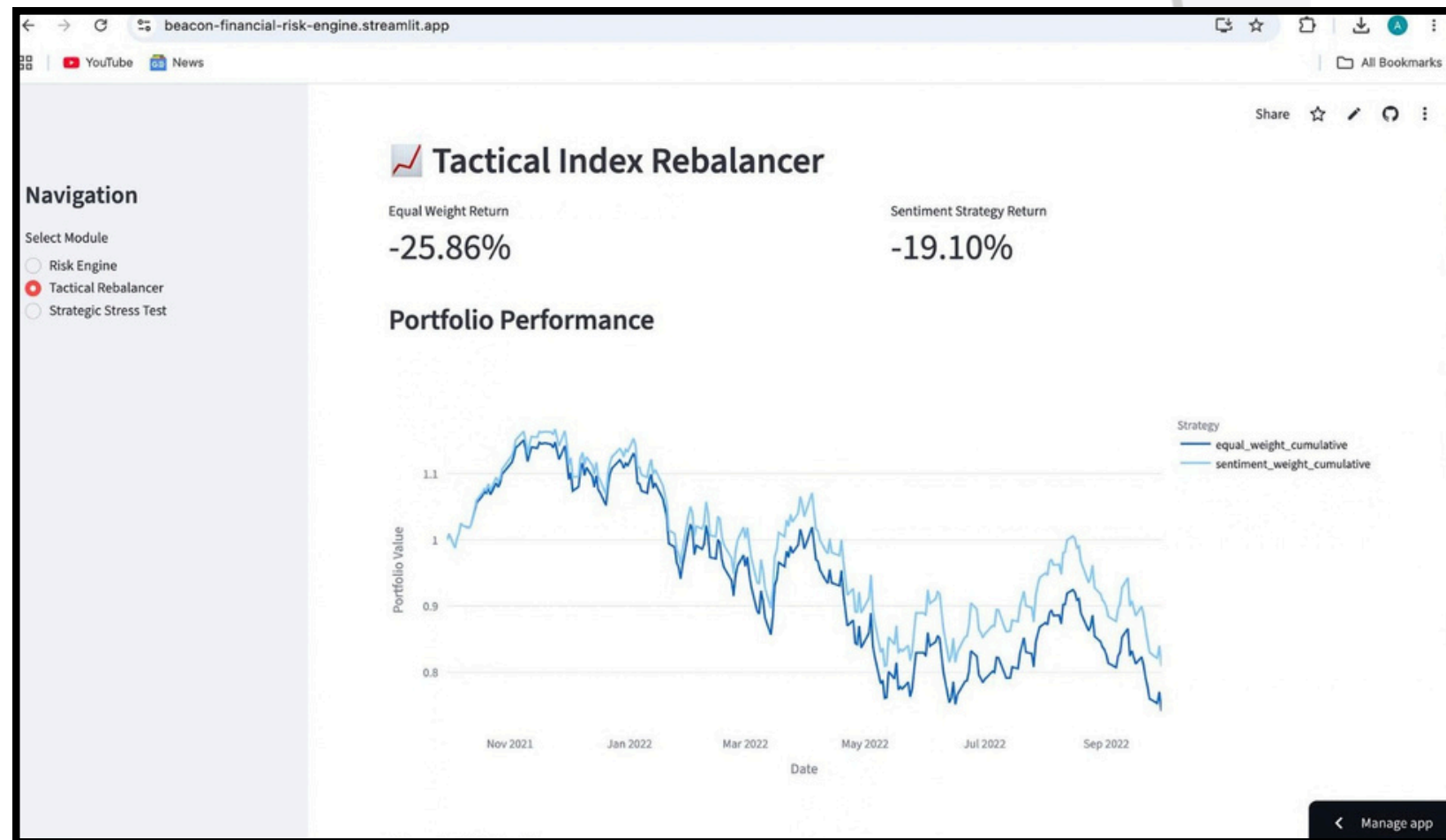
FinBERT Sentiment — **87.71% accuracy** | **0.87 macro F1**  
Event Classifier (TF-IDF + LR) — **72.34% accuracy** | **0.72 macro F1**

## Module A (Tactical)

Equal Weight return:  $-25.86\%$  vs Sentiment Strategy:  $-19.10\%$   
→ **+6.76 percentage-point improvement.**

## Module B (Stress Test)

Initial Portfolio: \$5,480.52M → Stressed Value: \$5,191.22M →  
Loss: \$289.30M (Impact:  $-5.28\%$ ).



# Domain Impact — Why Beacon Matters

- Rapidly processes large volumes of unstructured financial information into standardized signals.
- Automatically identifies events and estimates their potential market impact.
- Creates a common risk-signal layer that bridges research, trading, and risk teams.

For portfolio managers: sentiment supports tactical allocation; high-impact events trigger stress scenarios and pre-defined mitigation logic. Business value = faster information processing + structured risk intelligence + portfolio decision support.



## Current Limitations

- Event labels depend on weak supervision—label noise limits classifier ceiling.
- Impact score is an estimated heuristic, not yet trained directly on market moves.
- Datasets (news vs stock history) cover different time windows; Module A is a simplified historical backtest.
- Module B uses a synthetic portfolio and simplified stress assumptions—needs live validation.

## Planned Next Steps

1. Real-time news & social ingestion with robust deduplication and latency SLAs.
2. Fine-tune event classification with stronger supervision and weak-to-strong label pipelines.
3. Train direct text→market-impact models using aligned event/price datasets.
4. Deploy real-time risk alerts, SHAP explainability, and integrate true portfolio exposures for live rebalancing.

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Beacon transforms text into portfolio impact — from unstructured signals to actionable risk intelligence.